



School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Monthly Review – July 21, 2025 to August 21, 2025

TITLE OF THE PROJECT: Transfer Learning for Waste Classification and Reinforcement Learning for Sustainable Resource Optimization

STUDENT DETAILS:

S.No	VTU No	Register No	Name of the Student
1.	23403	22UEAD0040	J NAVADEEP
2.	23598	22UEAD0004	B SAI SUMANTH

ATTENDANCE & MEETING RECORDS:

S: No	Date	Task Updates	Student 1 Signature	Student 2 Signature	Student 3 Signature	Supervisor Signature
1	21/07/2025	Held the initial discussion with the supervisor to define the project scope. Identified objectives such as RL-based waste classification, sustainable resource tracking, and optimization of recycling pathways.				



2	22/07/2025	Reviewed multiple research papers on AI-based waste sorting, sustainable waste management frameworks, and circular economy strategies. Collected reference materials for methodology design.				
3	23/07/2025	Conducted a focused literature survey on waste classification frameworks for real-time detection.				
4	24/07/2025	Finalized the problem statement and project objectives. Drafted preliminary system architecture combining AI classification, cloud-based dashboards, and sustainability analytics.				
5	25/07/2025	Analyzed datasets available for waste type classification (organic, recyclable, hazardous, plastic categories, etc.). Shortlisted open-source image datasets for model training.				
6	26/07/2025	(Weekend) Studied existing waste management technologies and identified integration points where AI can enhance sorting accuracy and recycling efficiency.				
7	27/07/2025	(Weekend) Designed the high-level workflow: waste detection → classification by category → resource optimization → dashboard visualization.				
8	28/07/2025	Set up development environment (Python, TensorFlow, Streamlit dashboard, PostgreSQL). Created GitHub repository for version control and collaboration.				
9	29/07/2025	Pre-processed sample datasets and conducted exploratory data analysis to understand the class distribution of different waste categories.				



10	30/07/2025	Implemented a baseline ResNet50s model for waste image classification and performed initial training on the sample dataset.				
11	31/07/2025	Evaluated baseline model performance. Documented accuracy and limitations. Planned improvements using transfer learning				
12	01/08/2025	(Weekend) Studied sustainability dashboards and finalized key metrics (waste diverted from landfills, recycling efficiency, energy saved, CO ₂ reduction).				
13	04/08/2025	Designed database schema for waste traceability. Drafted data flow for mapping collected waste to households, industries, or urban sources.				
14	05/08/2025	Researched predictive modeling techniques for optimizing recycling efficiency. Noted ML regression approaches for resource allocation and processing optimization.				
15	08/08/2025	(Weekend) Conducted a mini-survey of existing waste management systems. Identified challenges and gaps that our AI-driven classification framework aims to solve.				



16	09/08/2025	Improved classification model using transfer learning. Achieved higher accuracy and enabled real-time detection for edge devices (smart bins/cameras).				
17	10/08/2025	Documented the current status of dataset acquisition, including open-source availability and pending discussions with potential providers.				
18	11/08/2025	Prepared draft of monthly progress report. Summarized literature review, system architecture, initial AI model results, and dashboard prototype.				
19	12/08/2025	Reviewed progress report and methodology outline with supervisor for feedback and validation.				
20	13/08/2025	Attended zeroth review meeting and presented current project status (awaiting dataset)				
21	23/08/2025	Attended zeroth review				

VERIFIED BY

(Signature of the Project coordinator-AIDS)

(Signature of the HoD-AIDS)